Class 9 Mathematics — Sindh Board

Exercise 4.1

Exercise 4.1

Factorize the following

(i), (ii), (iii), (iv), (v), (vi)

(i)

$$4x + 16y + 24z$$

Solution:

$$\text{Since, } ab + ac = a(b + c)$$

$$\text{Here, common factor} = 4$$

Answer: $4(x + 4y + 6z)$

(ii)

$$x^2 + 3x^2y + 4x^2y^2z$$

Solution:

$$\text{Since, } ab + ac = a(b + c)$$

$$\text{Here, common factor} = x^2$$

Answer: $x^2(1 + 3y + 4y^2z)$

(iii)

$$3pqr + 6pqt + 3pqs$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor $= 3pq$

Answer: $3pq(r + 2t + s)$

(iv)

$$9qr(s^2 + t^2) + 18q^2r^2(s^2 + t^2)$$

Solution:

Since, $ab + ac = a(b + c)$

Here, common factor $= 9qr(s^2 + t^2)$

Answer: $9qr(s^2 + t^2)(1 + 2qr)$

(v)

$$\frac{z^2x}{16}-\frac{x^2z^2}{8}+\frac{x^2z^3}{12}$$

Solution:

$$\text{Since, } ab+ac=a(b+c)$$

$$\text{Here, common factor}=\frac{xz^2}{48}$$

Answer: $\frac{xz^2}{48}(3-6x+4xz)$

(vi)

$$a(x-y)-a^2b(x-y)+a^2b^2(x-y)$$

Solution:

$$\text{Since, } ab+ac=a(b+c)$$

$$\text{Here, common factor}=a(x-y)$$

Answer: $a(x-y)(1-ab+ab^2)$

Exercise 4.1

Factorize the following

(i), (ii), (iii), (iv)

(i)

$$7x+xz+7z+z^2$$

Solution:

Since, $ax+ay+bx+by=(x+y)(a+b)$

$$(7x+xz)+(7z+z^2)$$

$$x(7+z)+z(7+z)$$

Answer: $(x+z)(z+7)$

(ii)

$$9a^2b+18ab^2-6ac-12bc$$

Solution:

Since, $ax+ay+bx+by=(x+y)(a+b)$

$$(9a^2b+18ab^2)-(6ac+12bc)$$

$$9ab(a+2b)-6c(a+2b)$$

Answer: $3(a+2b)(3ab-2c)$

(iii)

$$6t - 12p + 4tq - 8pq$$

Solution:

Since, $ax + ay + bx + by = (x + y)(a + b)$

$(6t - 12p) + (4tq - 8pq)$

$6(t - 2p) + 4q(t - 2p)$

Answer: $2(t - 2p)(2q + 3)$

(iv)

$$r^2 + 9rs - 7rs - 63s^2$$

Solution:

Since, $ax + ay + bx + by = (x + y)(a + b)$

$(r^2 + 9rs) - (7rs + 63s^2)$

$r(r + 9s) - 7s(r + 9s)$

Answer: $(r + 9s)(r - 7s)$

Exercise 4.1

Find the factors of

(i), (ii), (iii), (iv), (v), (vi)

(i)

$$4a^2 + 12ab + 9b^2$$

Solution:

Since, $a^2 + 2ab + b^2 = (a+b)^2$

Here, $a = 2a, \quad b = 3b$

Answer: $(2a+3b)^2$

(ii)

$$36x^4 + 12x^2 + 1$$

Solution:

Since, $a^2 + 2ab + b^2 = (a+b)^2$

Here, $a = 6x^2, \quad b = 1$

Answer: $(6x^2+1)^2$

(iii)

$$x^2 + 1 + \frac{1}{4x^2}$$

Solution:

$$\text{Since, } a^2 + 2ab + b^2 = (a+b)^2$$

$$\text{Here, } a = x, \quad b = \frac{1}{2x}$$

Answer: $\left(x + \frac{1}{2x}\right)^2$

(iv)

$$81y^2 + 144yz + 64z^2$$

Solution:

$$\text{Since, } a^2 + 2ab + b^2 = (a+b)^2$$

$$\text{Here, } a = 9y, \quad b = 8z$$

Answer: $(9y + 8z)^2$

(v)

$$625 + 50a^2b + a^4b^2$$

Solution:

$$\text{Since, } a^2 + 2ab + b^2 = (a+b)^2$$

$$\text{Here, } a = 25, \quad b = a^2b$$

Answer: $(25 + a^2b)^2$

(vi)

$$a^2 + 0.4a + 0.04$$

Solution:

$$\text{Since, } a^2 + 2ab + b^2 = (a+b)^2$$

$$\text{Here, } a = a, \quad b = 0.2$$

Answer: $(a + 0.2)^2$

Exercise 4.1

Factorize

(i), (ii), (iii), (iv), (v), (vi)

(i)

$$b^4 - 4b^2c^2 + 4c^4$$

Solution:

$$\text{Since, } a^2 - 2ab + b^2 = (a-b)^2$$

$$\text{Here, } a = b^2, \quad b = 2c^2$$

Answer: $(b^2 - 2c^2)^2$

(ii)

$$\frac{9}{4}x^4 - 2 + \frac{4}{9x^4}$$

Solution:

$$\text{Since, } a^2 - 2ab + b^2 = (a-b)^2$$

$$\text{Here, } a = \frac{3x^2}{2}, \quad b = \frac{2}{3x^2}$$

Answer: $\left(\frac{3x^2}{2} - \frac{2}{3x^2}\right)^2$

(iii)

$$2a^3b^3 - 16a^2b^4 + 32ab^5$$

Solution:

Since, $ab + ac = a(b+c)$

Here, common factor $= 2ab^3 \Rightarrow 2ab^3(a^2 - 8ab + 16b^2)$

Since, $a^2 - 2ab + b^2 = (a-b)^2 \Rightarrow a = a,\ b = 4b$

Answer: $2ab^3(a-4b)^2$

(iv)

$$9(p+q)^2 - 6(p+q)r^2 + r^4$$

Solution:

Since, $a^2 - 2ab + b^2 = (a-b)^2$

Here, $a = 3(p+q),\quad b = r^2$

Answer: $(3p + 3q - r^2)^2$

(v)

$$x^2y^2 - 0.1xy + 0.0025$$

Solution:

$$\text{Since, } a^2 - 2ab + b^2 = (a-b)^2$$

$$\text{Here, } a = xy, \quad b = 0.05$$

Answer: $(xy - 0.05)^2$

(vi)

$$(a-b)^2 - 18(a-b) + 81$$

Solution:

$$\text{Since, } a^2 - 2ab + b^2 = (a-b)^2$$

$$\text{Here, } a = (a-b), \quad b = 9$$

Answer: $(a - b - 9)^2$

Exercise 4.1

Factorize

(i), (ii), (iii), (iv), (v), (vi)

(i)

$$4a^2 - 9b^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a-b)(a+b)$$

$$\text{Here, } a = 2a, \quad b = 3b$$

Answer: $(2a - 3b)(2a + 3b)$

(ii)

$$16x^2 - 25y^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a-b)(a+b)$$

$$\text{Here, } a = 4x, \quad b = 5y$$

Answer: $(4x - 5y)(4x + 5y)$

(iii)

$$100x^2z^2 - y^4$$

Solution:

$$\text{Since, } a^2 - b^2 = (a-b)(a+b)$$

$$\text{Here, } a = 10xz, \quad b = y^2$$

Answer: $(10xz - y^2)(10xz + y^2)$

(iv)

$$\frac{1}{100}x^4 - 100y^4$$

Solution:

$$\text{Since, } a^2 - b^2 = (a-b)(a+b)$$

$$\text{Here, } a = \frac{x^2}{10}, \quad b = 10y^2$$

Answer: $\left(\frac{x^2}{10} - 10y^2\right)\left(\frac{x^2}{10} + 10y^2\right)$

(v)

$$\frac{64}{81}f^2-\frac{81}{64}g^4$$

Solution:

$$\text{Since, } a^2-b^2=(a-b)(a+b)$$

$$\text{Here, } a=\frac{8f}{9},\quad b=\frac{9g^2}{8}$$

Answer: $\left(\frac{8f}{9}-\frac{9g^2}{8}\right)\left(\frac{8f}{9}+\frac{9g^2}{8}\right)$

(vi)

$$\frac{x^4}{121}-121y^2$$

Solution:

$$\text{Since, } a^2-b^2=(a-b)(a+b)$$

$$\text{Here, } a=\frac{x^2}{11},\quad b=11y$$

Answer: $\left(\frac{x^2}{11}-11y\right)\left(\frac{x^2}{11}+11y\right)$

Exercise 4.1

Find the factors of

(i), (ii), (iii), (iv), (v), (vi)

(i)

$$(2x+z)^2-(2x-z)^2$$

Solution:

$$\text{Since, } a^2-b^2=(a-b)(a+b)$$

$$\text{Here, } a=(2x+z), \quad b=(2x-z)$$

$$a-b=2z, \quad a+b=4x$$

Answer: $8xz$

(ii)

$$(4a-9b)^2-(2a+5b)^2$$

Solution:

$$\text{Since, } a^2-b^2=(a-b)(a+b)$$

$$\text{Here, } a=(4a-9b), \quad b=(2a+5b)$$

$$a-b=2(a-7b), \quad a+b=2(3a-2b)$$

Answer: $4(a-7b)(3a-2b)$

(iii)

$$169x^4 - (3t+4)^2$$

Solution:

Since, $a^2 - b^2 = (a-b)(a+b)$

Here, $a = 13x^2, \quad b = (3t+4)$

Answer: $(13x^2 - 3t - 4)(13x^2 + 3t + 4)$

(iv)

$$(9x^2 - 4y^2)^2 - (4x^2 - y^2)^2$$

Solution:

Since, $a^2 - b^2 = (a-b)(a+b)$

Here, $a = (9x^2 - 4y^2), \quad b = (4x^2 - y^2)$

$a - b = 5x^2 - 3y^2, \quad a + b = 13x^2 - 5y^2$

Answer: $(5x^2 - 3y^2)(13x^2 - 5y^2)$

(v)

$$\left(a^2+2+\frac{1}{a^2}\right)-\left(b^2-2+\frac{1}{b^2}\right)$$

Solution:

Since, $a^2+2ab+b^2=(a+b)^2$ and $a^2-2ab+b^2=(a-b)^2$

$\left(a+\frac{1}{a}\right)^2-\left(b-\frac{1}{b}\right)^2$

Since, $a^2-b^2=(a-b)(a+b)$

Answer: $\left(a+\frac{1}{a}-b+\frac{1}{b}\right)\left(a+\frac{1}{a}+b-\frac{1}{b}\right)$

(vi)

$$9x^2+\frac{1}{9x^2}-4y^2-\frac{1}{4y^2}+4$$

Solution:

Since, $a^2+2ab+b^2=(a+b)^2$ and $a^2-2ab+b^2=(a-b)^2$

$\left(3x+\frac{1}{3x}\right)^2-\left(2y-\frac{1}{2y}\right)^2$

Since, $a^2-b^2=(a-b)(a+b)$

Answer: $\left(3x+\frac{1}{3x}-2y+\frac{1}{2y}\right)\left(3x+\frac{1}{3x}+2y-\frac{1}{2y}\right)$

Exercise 4.1

Find the factors of

(i), (ii)

(i)

$$(x^2+2xy+y^2)-9z^4$$

Solution:

$$\text{Since, } a^2+2ab+b^2=(a+b)^2$$

$$(x+y)^2-(3z^2)^2$$

$$\text{Since, } a^2-b^2=(a-b)(a+b)$$

Answer: $(x+y-3z^2)(x+y+3z^2)$

(ii)

$$(4a^2+8ab^2+4b^4)-9c^2$$

Solution:

$$\text{Since, } a^2+2ab+b^2=(a+b)^2$$

$$(2a+2b^2)^2-(3c)^2$$

$$\text{Since, } a^2-b^2=(a-b)(a+b)$$

Answer: $(2a+2b^2-3c)(2a+2b^2+3c)$